

Technology Stack

Date	06 July 2026
Team ID	
Project Name	Smart Lender — Applicant Credibility Prediction for Loan Approval
Maximum Marks	2 Marks

Technology Stack Details

The technologies below were selected for being lightweight, well-documented, and directly suited to a small ML web application with a five-person student team.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 templates	Server-rendered pages keep the stack simple; no build tooling needed for a small form-based UI
2	Backend / Server-Side	Python, Flask	Lightweight web framework that integrates directly with scikit-learn/XGBoost objects already in Python
3	Database / Data Storage	CSV file (loan_prediction.csv); pickled/JSON model artifacts	Dataset size (614 rows) does not require a full RDBMS; file-based storage is sufficient for this scope
4	Cloud / Hosting / Deployment	Local Flask dev server (Anaconda); IBM Cloud per architecture design	Local testing for development; IBM Cloud specified in the original architecture for production deployment
5	Version Control & CI/CD	Git, GitHub	Enables team collaboration on shared code and a single submission repository
6	Third-Party APIs / Other Tools	scikit-learn, XGBoost, imbalanced-learn (SMOTE), pandas, NumPy, Matplotlib, Seaborn	Core ML/data libraries used for preprocessing, modeling, and visualization in the notebook